

1. Abstract:

Mass spectrometry (MS)-based proteomics requires accurate protein databases to be used in comparison to experimental data, which tend to be unavailable for newly sequenced organisms. To address the requirement of accurate protein prediction in these cases, a python-based proteomics pipeline was created which can predict the possible proteins from genomic sequences, digests them with one of four proteases (Trypsin, Lys-C, Arg-C, and Glu-C), calculates m/z values and infers ion statistics which aid in evaluating protease efficiency based on unique peptide identification. Out of the four enzymes analysed by the pipeline, Trypsin produced the highest unique peptides, within the 1000-1500 m/z range. While the other enzymes offered less unique peptide identification, they can offer alternative coverage of the data to complement the digestion by Trypsin. Overall, this pipeline provides an efficient strategy for protease evaluation and database generation when working with lesser characterized organisms.

2. Introduction:

A difficulty researchers face when sequencing newly discovered organisms is the lack of protein annotation of the organism, which in turn limits protein identification by mass spectrometry (MS). MS-based proteomics depends primarily on comparing the experimental data against a reference protein database, and the absence of a complete and accurate database for newly sequenced or non-model organisms raises the chance of misidentified taxa (Alfredo Blakeley-Ruiz & Kleiner, 2022). This project tackles this issue by first predicting all possible proteins produced by the organism. It then tests different proteases and evaluates which method produces the most identifiable proteins.

The pipeline applies the logic of bottom-up or shotgun proteomics, a well-established approach where small peptides are digested and analyzed by MS to identify and quantify the present proteins (Chevalier et al., 2015), *in silico*. It starts with computationally predicting all possible protein sequences to create a reference database, which is then digested to mimic experimental enzymatic treatments. To ensure high protein coverage, eliminate Trypsin bias, and quantify proteins accurately four different proteases (Trypsin, Lys-C, Arg-C, and Glu-C) were used as recommended by Woessmann et al. (2022), while considering missed cleavages.

Since the principle of MS depends on the movement of charged ions inside an electromagnetic field, the results are converted to mass-to-charge values (m/z) by ionizing the peptides to be measured (Prabhakar Babele & Amit Kumar Yadav, 2023). The m/z values are processed to perform statistical analyses that include counting peptides within specified mass ranges, creating binned histograms, and identifying unique proteins based on m/z values. The aim of this project is to evaluate the enzyme with the most uniquely identified peptides.

3. Methods:

3.1 ORF finder:

The program examines six frames, three forward and three reverse (computed from the reverse complement). Start and stop codons were identified within the frames to predict the ORFs, with a minimum length of 50 amino acids to reduce false positives (Beals et al., 2024), disregarding internally nested ORFs and ORFs that contain ambiguously called bases. The ORF finder then reports the translated ORFs in FASTA format.

3.2 Protein Digester:

The digester has selectable proteases from the following list Trypsin: cuts at K or R, Endoproteinase Lys-C: cuts at K, Endoproteinase Arg-C: cuts at R, V8 proteinase (Glu-C): cuts at E. None of these proteases restricts if the amino acid is followed by a Proline. The protein digester then reports the individual peptides post-protease restriction in FASTA format.

3.3 Mass Analyzer:

The mass analyzer converts peptide sequences into mass-to-charge (m/z) values to four decimal places, using either average or monoisotopic amino acid mass. The program reports the name, peptide number, m/z , charge, missed cleavage sites, and the peptide sequences in tab separated value (TSV) format.

3.4 Ion statistic calculator:

The ion statistic calculator has several modes: mode 1 reports the number of peptides within a mass range, mode 2 calculates the number of peptides based on user provided m/z bins, mode 3 performs similar calculations to mode 2 but with scalable sliding windows, and finally mode 4 determines the number of unique peptides to a protein and thus how many proteins have been identified. These data are reported as a CSV file.

3.5 Terminal command to run entire pipeline:

```
python3 group2_task1.py --input genome.fasta --output output1.fasta && python3
group2_task2.py -i output1.fasta -o output2.peps -e 2 -m 1 && python3 group2_task3.py
output2.peps output3.pepmasses m && python3 group2_task4.py output3.pepmasses -m 1 &&
python3 group2_task4.py output3.pepmasses -m 2 -b 1.0 && python3 group2_task4.py
output3.pepmasses -m 3 --window 200 --step 50 && python3 group2_task4.py
output3.pepmasses -m 4
```

4. Results:

Using the peptide mass files generated from the workflow, the total number of peptides and uniquely identified proteins were measured for each enzyme, alongside their m/z distribution profiles. The combined table and histogram illustrate the clear differences in digestion performance with Trypsin yielding the highest peptide coverage within the 1000-1500 m/z range.

Table: Comparison of the total number of peptides, uniquely identified peptide masses, and the uniquely identified proteins produced by the four different types of proteolytic enzymes. Trypsin

produces the highest number of uniquely identified proteins whereas V8 proteinase produces the lowest.

Enzyme	Total Peptides	Uniquely identified peptide masses	Uniquely identified proteins
<i>Trypsin</i>	14847	11992	1927
<i>Endoproteinase Lys-C</i>	6401	5765	1629
<i>Endoproteinase Arg-C</i>	4296	4018	1428
<i>V8 proteinase</i>	6200	5524	1225

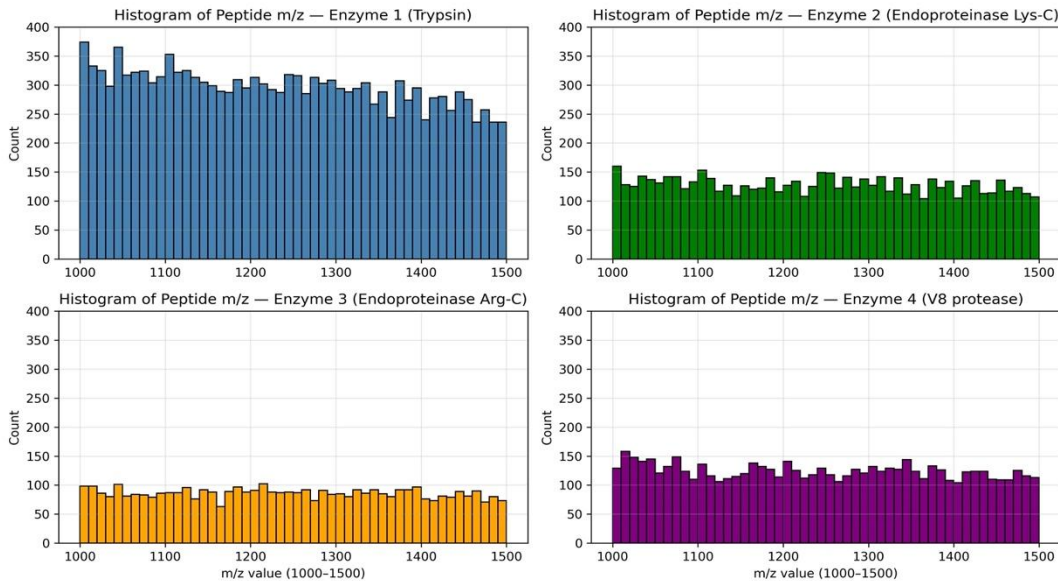


Figure1: Histogram plots of the peptide m/z values (1000 – 1500 Da) generated for all four enzymes (Trypsin, Endoproteinase Lys-C, Endoproteinase Arg-C, V8 protease). The distribution highlights how different proteases yield distinct peptide mass profiles within the selected m/z range.

5. Discussion:

The comparison of the four digestive enzymes after digestion and mass analysis using the pipeline revealed that trypsin is the best performing among the four enzymes to use when identifying peptides and unique proteins in a MS-based analysis. All remaining three enzymes identified fewer peptides and unique proteins compared to trypsin. This finding is consistent with the trypsin's well characterized advantages in MS-based proteomics (Burkhart et al., 2012).

The pipeline effectively performs its purpose of determining the most effective protease in identifying unique peptides as trypsin. However, utilizing other enzymes in combination with trypsin, or as an alternative to trypsin, can help increase the efficiency of the experiment and reduce the protease-based bias (Tran, 2018).

References

- Alfredo Blakeley-Ruiz, J., & Kleiner, M. (2022). Considerations for Constructing a Protein Sequence Database for Metaproteomics. *Computational and Structural Biotechnology Journal*. <https://doi.org/10.1016/j.csbj.2022.01.018>
- Beals, J., Hu, H., & Li, X. (2024). A survey of experimental and computational identification of small proteins. *Briefings in Bioinformatics*, 25(4). <https://doi.org/10.1093/bib/bbae345>
- Burkhardt, J. M., Schumbrutski, C., Wortelkamp, S., Sickmann, A., & Zahedi, R. P. (2012). Systematic and quantitative comparison of digest efficiency and specificity reveals the impact of trypsin quality on MS-based proteomics. *Journal of Proteomics*, 75(4), 1454–1462. <https://doi.org/10.1016/j.jprot.2011.11.016>
- Chevalier, F., Hamdi, D. H., Saintigny, Y., & Lefaix, J.-L. (2015). Proteomic overview and perspectives of the radiation-induced bystander effects. *Mutation Research/Reviews in Mutation Research*, 763, 280–293. <https://doi.org/10.1016/j.mrrev.2014.11.008>
- Prabhakar Babele, & Amit Kumar Yadav. (2023). Back2Basics: mass-to-charge ratio (m/z) in proteomics. *Journal of Proteins and Proteomics*, 14(4), 223–226. <https://doi.org/10.1007/s42485-023-00115-7>
- Tran, D. T. (2018). Engineering Proteases for Mass Spectrometry-Based Post Translational Modification Analyses. *PROTEOMICS*, 19(10), 1700471. <https://doi.org/10.1002/pmic.201700471>
- Woessmann, J., Kotol, D., Hober, A., Uhlén, M., & Edfors, F. (2022). Addressing the Protease Bias in Quantitative Proteomics. *Journal of Proteome Research*, 21(10), 2526–2534. <https://doi.org/10.1021/acs.jproteome.2c00491>